A Bi-modal Automated Essay Scoring System for Handwritten Essays

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***Abstract*—During the past few decades, Automated Essay Scoring (AES) technology has been widely used to alleviate the workload of teachers and improve the feedback cycle in educational systems. However, the scoring of handwritten essays poses great challenges for existing systems, since most of them only take text as input without consideration of errors or bias which may be introduced by Optical Character Recognition (OCR) as a necessary pre-processing step. This paper proposes VisualAES, a bi-modal automated essay scoring system that utilizes both textual and visual features for handwritten essay scoring. Specifically, we first employ three powerful pre-trained transformer-based models as the backbone, and extend them to take both textual features and visual features extracted by Faster R-CNN. Then, a stacking ensemble model is subsequently adopted to robustly map their outputs to a final score. We evaluate VisualAES with the public Automated Student Assessment Prize (ASAP) dataset and our proposed handwritten Chinese Students' handwritten essay dataset (ChnStd). Results show that the proposed VisualAES outperforms all state-of-the-art methods on both datasets. More importantly, by incorporating handwritten image information, we also achieve a further performance improvement on ChnStd and reduce the side effects of OCR.**

***Index Terms*—bi-modal, transformer, AES, handwritten**

I. Introduction

As one of the most important educational applications of natural language processing (NLP), automated essay scoring (AES) has been widely employed to reduce the bias of human raters and increase marking efficiency. Many well-known English tests have proven the validity and feasibility of modern AES systems [1], such as Graduate Record Examination (GRE) and Test of English as a Foreign Language (TOEFL). There are also many middle and high schools across the world that adopt AES for end-of-year standardized assessments or entrance exams. In practice, AES models are trained on a small set of essays scored by hand, and then score new essays with the reliability of expert raters [2]. A great amount of work during last few decade has demonstrated this reliability [3].

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In many countries the major testing form is still written examination on papers rather than computer-based testing. For instance, most high school and college entrance examinations are paper-based in developing countries. To use existing AES systems for handwritten essays, it's inevitable to adopt Optical Character Recognition (OCR) as a pre-processing step. OCR is the electronic or mechanical conversion of images of typed, handwritten or printed text into machine-encoded text. By adopting OCR, the handwritten images can be recognized as text inputs for AES. But, OCR could introduce an extra loss because of inevitable errors during the recognition process.

Therefore, it is imperative to incorporate both text and handwritten information for AES. Recent advances in deep learning empower impressive performance and functionality improvements by integrating the ideas of both NLP and Computer Vision (CV), which also bring us inspiration and potential to enhance AES with these advances. For example, the attention mechanism of the translation task is introduced into classic image classification, and achieves promising results [4]. The Convolutional Neural Network (CNN) architecture is also applied to encode character writing for language model [5]. The fusion of them has further stimulated more interesting research topics, e.g. Image-to-Text [6], and Text-to-image tasks [7], visual question answering [8] and so on. These research absorb advantages from both NLP and CV fields, and lays a solid foundation for related fields.

We propose VisualAES, a bi-modal automated essay scoring system that accepts both text and corresponding handwritten essay images as inputs for handwritten essay scoring. To our best knowledge, this is the first bi-modal AES system incorporating handwritten images as an extra input. Particularly, we first adopt Faster R-CNN [9] to extract visual features from handwritten images, which are then concatenated with word embedding obtained by a pre-trained transformer encoder. These features are integrally fed into three modified transformer based models (i.e. Visual-Bert, Visual-RoBerta and Visual-Albert) for bi-modal scoring. Finally, a stacking

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ensemble module is trained to yield a final heterogeneous score using predictions from the three base models. In this way, it is able to score an essay with regard to extra image information and reduce performance degradation caused by OCR loss by introducing visual modality. Evaluated with ASAP [10] dataset and our proposed Chinese Students' handwritten essay dataset(ChnStd), the proposed system shows superior performance compared with all existing methods.

This paper contributes in the following three aspects:

- We propose the first bi-modal AES system, which applies both original image and corresponding text features to score handwritten essays.
- We build and public a bi-modal Chinese Students' handwritten essay dataset (ChnStd), which consists of both original handwritten images and corresponding expert-checked textual files. Experiments demonstrate that the proposed VisualAES achieves state-of-the-art performance on both ASAP and ChnStd datasets.
- We prove the effectiveness of incorporating visual handwritten features by comparing single modal with bi-modal scoring performance. We also demonstrate that bi-modal scoring is able to reduce the negative effect caused by OCR errors and improve the generalization capability.

II. Related Work

In this section, we first present an overview of the major milestones in AES research, and then discuss most relevant tasks and progresses in related NLP and CV fields, including image feature extractor and text-image fusion.

A. Existing AES systems

Virtually all current AES systems are learning-based and can be classified based on the form of learning algorithms they employ (e.g. supervised, semi-supervised, or reinforcement learning). We will focus on supervised applications of AES in this subsection, as state-of-the-art systems are all supervised and refer the reader to [11] for approach of weakly supervised learning or [12] for the approach of reinforcement learning to AES.

There are numerous works on supervised learning AES applications which focus on how to model text input to score essays automatically.

Recurrent Neural Networks (RNN) [13] and Convolutional Neural Networks (CNN) [14] have been used for modelling input essays [15]. Followed with [16], who use convolution layer to extract n-gram level features from word sequence in an essay and then apply a single-layer Long-Short Term Memory (LSTM) [17] network to predict the essay's score.

Reference [18] notice that some words contribute slightly in discriminating between good and bad essays and fail to distinguish these under informative words leads to performance loss. So the score-specific word embeddings (SSWE), which is a low-dimensional real-valued word vector representation, is trained and then fed into two-layer bi-directional LSTM (2L-BLSTM) to reduce the loss. Reference [19] combine low-level feature named Histogram Intersection String kernels (HISK) with high-level feature named the bag-of-super-word-embeddings (BOSWE) and employ LibSVM to score the essay. HISK captures the similarity among strings based on counting common character ngrams, and BOSWE is for semantic representation. Reference [20] incorporates an attention mechanism into the CNN by using attention pooling instead of simple pooling layer after layer because some characters, words and sentences in an essay are more important than the others. By leveraging attention mechanism, they obtains better performance than former systems. Reference [21] aims to score essays prompt-independent settings, thus a two-stage deep neural network(TDNN) is proposed. In the first stage, using the rated essays for non-target prompts as the training data, a shallow model is learned to select essays with an extreme quality for the target prompt, serving as pseudo training data; in the second stage, an end-to-end hybrid deep model is proposed to learn a prompt-dependent rating model consuming the pseudo training data from the first step.

B. Image Encoding

Image encoding refers to models that could extract information from images for different tasks. That is the most notably classic CNN architecture [4] in image classification task and its variants, e.g. AlexNet [22], VGGNet [23] and GoogleNet [24].

To solve the vanishing gradient problem in CNNs, ResNet is proposed by [25]. The main idea is inspired by LSTM which uses gate functions to control or pass-through the neuron's activation value.

As an important concept of CNN, pooling lowers the computational burden by reducing the connection number between layers. To improve the pooling layer, Reference [26] proposes the Spatial Pyramid Pooling (SPP), also known as DeepLab, which generates a fixed-length representation regardless of the size of the input. By replacing the pooling layer with SPP, the model could deal with images with different sizes.

C. Text-Image Fusion

There is a long research history of combining text and image features. Understanding detailed semantics depicted in an image is critical for visual understanding [27], and prior studies show that modeling such semantics can benefit visual-and-language models. Previously, task-specific models were developed, wherein the features derived from off-the-shelf computer vision and combined with NLP models in an ad-hoc way. And model training is performed on the dataset for the specific task only. Plenty of Vision-and-Language tasks like textual grounding [28], question answering [8], and visual reasoning [29] have been proposed, and various models [30], [31] have been designed to solve them. These models often consist of a text encoder, an image feature extractor, a multi-modal fusion module, and an answer classifier designed for specific tasks.

Inspired by BERT [32], a transformer-based representation model for natural language, VisualBERT [33] can be easily applied to new tasks with good generalization capability. It consists of a stack of transformer layers that implicitly align elements of an input text and regions in an associated input image with self-attention. To adapt to different tasks, VisualBERT is pre-trained with a masked language modeling and sentence-image prediction task on caption data and then fine-tuned for the other tasks.

Concurrent to VisualBERT, there are many models seeking to conduct pre-training for visual-linguistic tasks. Like VideoBERT [34] which transforms a video into a series of images paired with spoken words and learns joint representations using a transformer. Also, ViLBERT [35] proposes to learn joint representation of images and text by extending a BERT-like architecture to a multi-modal two-stream model. This model processes both visual and textual inputs in separate streams that interact through co-attentional transformer layers, but it results in twice the parameters. At the same time, [36] design VL-BERT, which adopts the simple but powerful transformer based model as the backbone, and extends it to use both visual and linguistic embedded features as input. In this model, each element of the input is either a word from the input sentence, or a region-of-interest (RoI) from the input image.

III. Method

This section details the proposed bi-modal VisualAES system, which consists of input text embedding, image feature embedding, text-image fusion, a series of extended transformer based models that accept joint representations of text and image for bi-modal scoring, and a stacking ensemble module that learns how to best combine the predictions from these models.

As shown in Figure 1, the original handwritten images provide both text and image content representations. The upper way utilizes a standard OCR model (CNN+LSTM+CTC) depicted by [37] to extract pure text content and then transforms to word embedding by a pre-trained vocabulary. The bottom way employs Faster R-CNN to extract image features. Then, they are concatenated to form our joint representations of bi-modal data.

A. Text Encoding

We use a pre-trained transformer encoder as the text encoder to process the input text. Similar to other sequence models, the input text is represented by tokens at first and then converted to vectors by looking up a mixture embedding. The mixture embedding is the sum of two parts: Xavier initialized embedding [38] and position embedding [39], which are computed as following:

$$
\begin{aligned}
&Emb_x = U[-\frac{\sqrt{6}}{\sqrt{t+d}}, \frac{\sqrt{6}}{\sqrt{t+d}}] \\
&Emb(pos) = \begin{cases} cos(\frac{pos}{10000^{i/d}}) + Emb_x, \\ \text{if } i\%2 = 0; \\ sin(\frac{pos}{10000^{i/d}}) + Emb_x, \\ \text{if } i\%2 = 1; \end{cases}
\end{aligned} \tag{1}
$$

where the t is the number of tokens; d represents the length of embedding; i refers to the dimension of embedding; pos is the position of tokens in sequence; Emb_x represents the Xavier initialized embedding. Then, word tokens are looked up in this mixture embedding and fed to a transformer with multi-head self-attention. The text encoder outputs textual features as f_{txt}, and d_{en} is the textual dimension of the pre-trained encoder.

Another part of text input is the textual tag. The tag encoding informs the type of information, we use a type token "txt" to uniformly represent textual information type. In order to inject tag information into f_{txt}, additional encoding layers are added. The additional layer is trainable to enable models to learn the tag feature, and the additional layer is modeled to have the same dimension d_{en} of the pre-trained textual encoder.

Then, we combine textual features with textual tags through element-wise summation, resulting in a rich text representation $feature_{txt}$ of length L_{txt}. L_{txt} represents the number of words in an essay.

B. Image Feature Extractor

We employ a modified Faster R-CNN to extract image features, in which we limit the location and size of each bounding box to fixed values during training as the nature of handwriting in the scanned image, i.e. one with 300×300 and the other four with 150×150, respectively. The parameters of Faster R-CNN are trainable and can be learned from various datasets.

$$f_{img}, fp_{img} = FasterR - CNN(I_{img}) \tag{2}$$

where f_{img} refers to the whole input image, and fp_{img} refers to the local features which represents each part of the image.

Finally, the outputs of Faster R-CNN is transformed to an image feature vector with the same dimension of text embedding, using a fully connected network (FCN) layer as follows:

$$feature_{img} = FCN(f_{img}, fp_{img}) \tag{3}$$

where the $feature_{img}$ refers to the final feature of an input image of length L_{img}. L_{img} represents the number of boxes, $L_{img} = 1 + 4$.

Similar to the textual tag feature, we use a type token "img" to represent image information uniformly and add the image tag to image representation.

Then, we concatenate the image feature $feature_{img}$ and its text feature $feature_{txt}$ to create a single input X_{de} of length $(L_{txt} + L_{img})$ and dimension d_{en}. The concatenation is utilized only if the token exists and not padded. Finally, we obtain the text-image joint representations of handwritten images to facilitate subsequential ensemble modal scoring.

C. Extended Transformer Based Models

Inspired by VL-BERT [36] that modifies the original BERT [32] by just adding new visual elements to input layers, and adopts the same transformer based backbone for visual-linguistic tasks, we extend two other powerful transformer

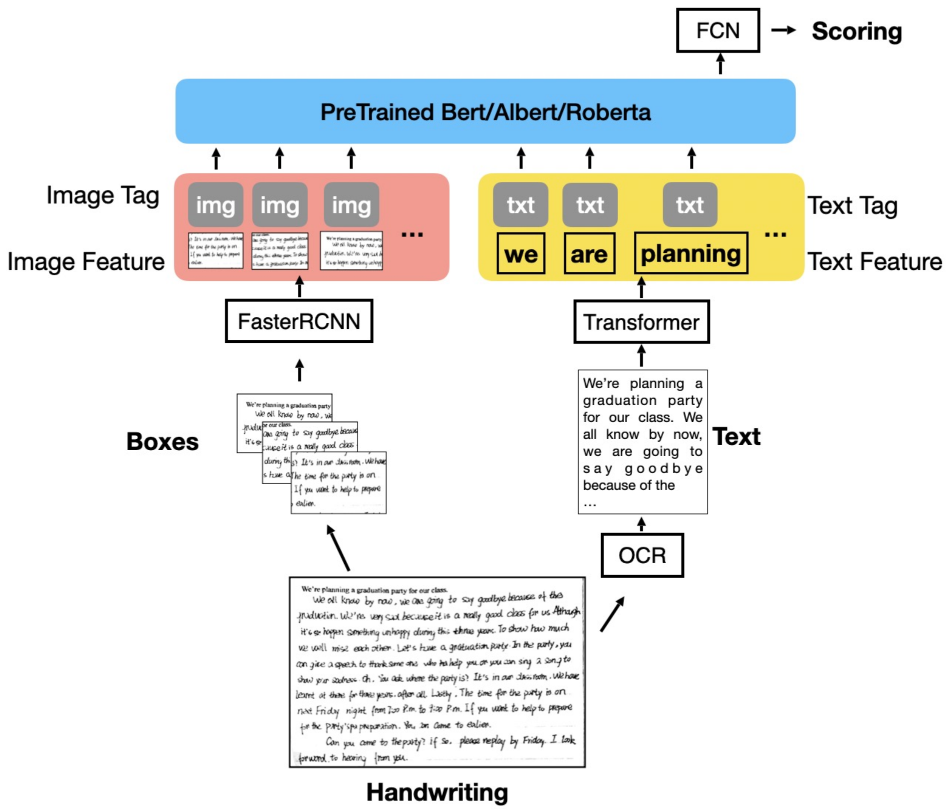


Fig. 1. Text-image fusion mechanism of VisualAES

based models, i.e. RoBerta [40] and AlBert [41] to accept both text and image embedded features for bi-modal handwritten essay scoring. The weights are pre-trained, and hyper parameters are inherited from default settings. The mask is applied to both word padding and image feature padding to avoid the meaningless gradients in propagation. Note that we did not attach visual geometry embedding of original VL-BERT to the end of text features in our method, since the bounding boxes are fixed and the regions-of-interest (RoIs) are the whole handwritten area.

Thus, we get three extended transformer based models, i.e. Visual-Bert, Visual-RoBerta, and Visual-AlBert, and their results are then fed into a stacking ensemble module for better predictive performance.

D. Stacking Ensemble Module

Ensemble learning is the process by which multiple models are strategically combined to achieve higher accuracy than that of each individual model and reduce the over-fitting risk. In VisualAES, we apply stacking ensemble that uses predictions from the three transformer-based models (i.e. Visual-Bert, Visual-RoBerta, and Visual-AlBert) to build a new model for a final heterogeneous score, as shown in Figure 2.

In detail, the three base models are trained based on the complete same training set as usual, and then we apply a

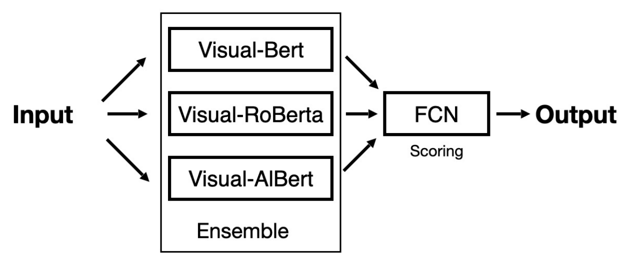


Fig. 2. The structure of VisualAES

FCN that is trained on the outputs of the three base models as features to generate the final score:

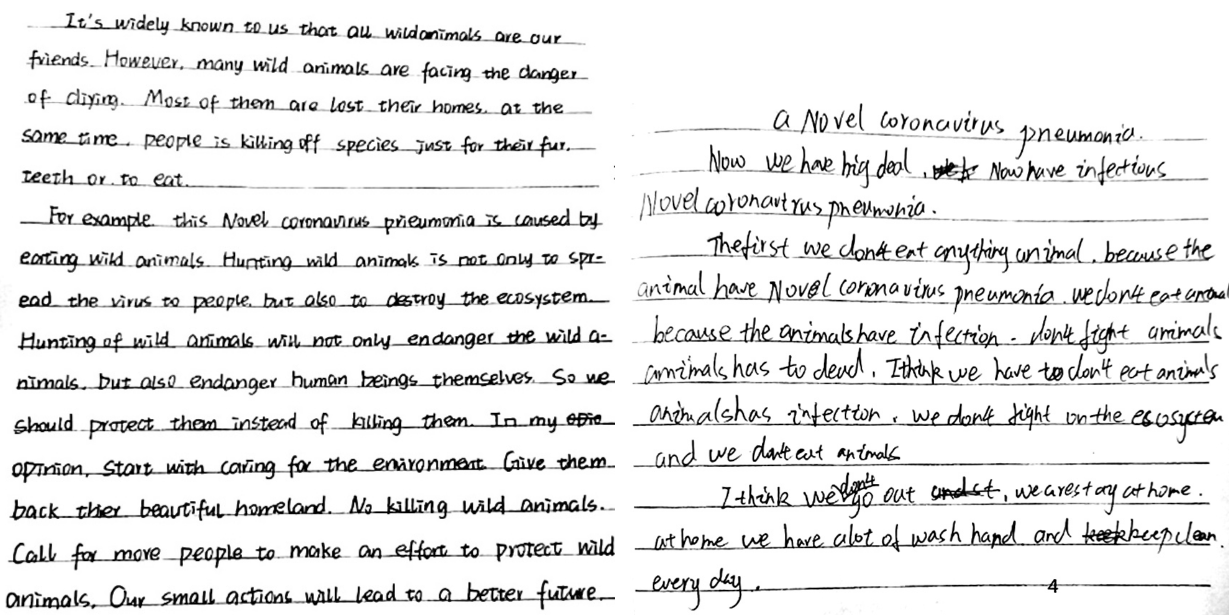
$$\begin{aligned} VisualAES = Ensemble(&Visual\_bert, \\ &Visual\_roberta, Visual\_albert) \end{aligned} \tag{4}$$

To accelerate the computing process, we employ three GPUs to parallelly perform each base model.

IV. Experiment

In experiment, we evaluate VisualAES on both the public single-modal dataset ASAP and our proposed bi-modal dataset ChnStd. Detailed dataset statistics and evaluation metrics are depicted in Section IV-A and Section IV-B.

Firstly, we compare our method with several state-of-the-art single-modal methods on ASAP and the textual part of


It's widely known to us that all wildanimals are our friends. However, many wild animals are facing the danger of diying. Most of them are lost their homes. at the same time. people is killing off species just for their fur, teeth or. to eat.

For example this Novel coronavirus prieumonia is caused by eating wild animals. Hunting wild animals is not only to spread the virus to people. but also to destroy the ecosystem. Hunting of wild animals will not only endanger the wild animals. but also endanger human beings themselves. So we should protect them instead of killing them. In my ~~opio~~ opinion, Start with caring for the environment. Give them back thier beautiful homeland. No killing wild animals. Call for more people to make an effort to protect wild animals. Our small actions will lead to a better future.

a Novel coronavirus pneumonia.

Now we have big deal, ~~we~~ Now have infections Novel coronavirus pneumonia.

Thefirst we don't eat anything animal. because the animal have Novel coronavirus pneumonia. we don't eat animals because the animals have infection. don't fight animals animals has to dead. Ithink we have to don't eat animal's animalshas infection. we don't fight on the ecosystem and we dont eat animals

I think we don't go out ~~andst~~, we arestay at home. at home we have alot of wash hand and ~~keek~~ keep clean every day.

4

Fig. 3. The real students' handwritten images. Left essay with less OCR recognition errors comparing with the right essay.

ChnStd in Section IV-C and Section IV-D, respectively. Then, we demonstrate the effectiveness of bi-modal VisualAES on the ChnStd dataset by incorporating both text and handwritten images in Section IV-E. As an indispensable pre-processing step for handwritten AES, OCR may introduce extra loss caused by automatic recognition errors. Therefore, we also evaluate VisualAES using both raw and human corrected OCR results to discover the influence of OCR recognition errors in Section IV-F. Moreover, in Section IV-G we evaluate the generalization capability of VisualAES via a blind testing, in which the essay topics and students of testing data are totally different from training. Note that as the ASAP provides only textual files without handwritten images, we test bi-modal VisualAES with blank images.

The initializing Faster R-CNN parameters used for VisualAES are inherited from VL-BERT, and the resolution of input images is set to 300×300. The backbone of Faster R-CNN is ResNet101. The dimension of word embedding and sequence length of transformers are 768 and 512, respectively. We recommend readers of VL-BERT [36] to get exhaustive parameters. We use a fully connected network (FCN) to ensemble the outputs of the three base models to get the final score.

A. Dataset Statistic

The ASAP dataset is collected from US students and consists of 8 different prompts of genres with three essay types (Argumentative, Response and Narrative). It has been widely used to test AES systems developed for native English speakers' essays. However, even though grammar or morphology are the same for both native and non-native speakers, non-native English raters commonly concentrate on language correctness while native English raters tend to evaluate logic or expressions of essays. Therefore, we propose a new bi-modal non-native speakers' dataset - ChnStd for a more comprehensive evaluation.

The ChnStd dataset contains 2600+ handwritten essays with three types of essays (Narrative, Suggestion and Letter) collected from Chinese middle school's English examinations in 2020, examples are shown in Figure 3. All testers are from Grade 7 to 9, with various levels of English writing capabilities. There are 45 teachers as the raters with at least 10 years teaching experiences. In order to make the score more objective, each essay is rated by 3 teachers at first, and the score range is 0-20. If the score difference is above 3 between any two of these three teachers, then additional teachers will be involved to score this essay until any three teachers reach an agreement, which means the score difference is less than 3. Generally, the final score of each essay is the average of three teachers' scores at least. We apply a standard OCR model (CNN+LSTM+CTC) [37] to produce corresponding pure textual files that contain word sequences of the essays. All the textual files are carefully corrected by teachers and then tagged with the same labels of corresponding images. The overall statistics of the dataset is shown in Table I. We also share our ChnStd[1] dataset to other researchers.

In Section IV-C to Section IV-F, we apply a randomly split process for evaluation, i.e. the training dataset accounts for 80 percent and testing dataset accounts for the rest 20 percent. In Section IV-G, we then train the model using Grade 7-8 dataset and perform testing on Grade 9 dataset to evaluate the generalization capability.

[1]https://pan.baidu.com/s/15bQ-IfoiKfOzSQL9nBneOA

TABLE I
THE STATISTIC OF ASAP AND CHNSTD DATASET. ESSAY TYPE INCLUDES: ARGUMENTATIVE (A), RESPONSE (R), NARRATIVE (N), COMMENT (C), SUGGESTION (S) AND LETTER (L).

Corpora	Essay Types	Language Level	No. of Essays	Scoring Task	Scoring Range	Image
ASAP	A, R, N	US Grades 7 to10	17450	Holistic	as small as 0-3, as large as 0-60	No
ChnStd	N, S, L	CN Grades 7 to 9	2632	Holistic	0-20	Yes

B. Evaluation Metrics

Since Kappa is a widely used metrics in previous work, we first evaluate the proposed system using the Cohen's Kappa with quadratic weights, i.e. Quadratic Weighted Kappa (QWK) [42], which takes into account the distance between the actual score and the reported score. However, the Kappa is capable for evaluating discrete values while the scores in the ChnStd dataset are continuous. Hence we introduce Pearson as the other evaluation metric, which provides the measure of linear correlation between ground truth and predicted scores. In this way, we could evaluate both discrete and continuous scores adequately. In experiments, the performances of models in ASAP dataset are evaluated in same one whole evaluation dataset without prompt separation (ASAP contains 8 prompts). That will boost the Kappa and Pearson value comparing with previous works. Mixing the prompts, we could test the ability of identifying the essay type and student grade by the model itself automatically.

C. Single-modal Native Speaker Scoring

We first compare our VisualAES with several baseline and state-of-the-art single-modal methods on the ASAP dataset, including RNN [13], LSTM [17], CNN [4], CNN+LSTM [43], 2L-LSTM [21], TDNN [21], and HISK+BOSWE [19]. The results are shown in Table II. In addition, we also involve three recent pre-trained models, i.e. Bert [32], Roberta [40], Albert [41] for a more comprehensive evaluation. All the testing is conducted without additional images fairly.

TABLE II
THE EVALUATION RESULTS OF DIFFERENT METHODS ON ASAP DATASET.

Model	Kappa	Pearson
RNN	0.745	-
LSTM	0.746	-
ML-LSTM(L=3)	0.788	-
CNN	0.808	-
CNN+LSTM	0.861	-
HISK+BOSWE	0.885	-
2L-LSTM	0.568	0.754
TDNN	0.786	0.824
Bert-base	0.974	0.977
Bert-large	0.981	0.983
Roberta-base	0.801	0.839
Roberta-large	0.981	0.984
Albert-base	0.981	0.979
Albert-large	0.980	0.982
VisualAES	**0.983**	**0.984**

Enhanced by stacking ensemble, the proposed VisualAES achieves the best performance among all methods in terms of both Kappa and Pearson, in spite of absence of image inputs. The improvements from single transformer-based models (Bert, Roberta, Albert) to VisualAES result from the ensemble methods. That explains the VisualAES could reach the best Pearson value 0.984 and get minor 0.02 improvement in Kappa value, comparing single transformer-based models to VisualAES. Besides, most pre-trained models also achieve impressive improvements in comparison to earlier neural network models like CNN, LSTM, etc. With the capability to extract higher-order features, these methods can better identify complex patterns within and across text sequences with considerably less human efforts.

D. Single-modal Non-native Speaker Scoring

We then evaluate VisualAES on the proposed ChnStd dataset. The input text is carefully checked by human experts to avoid OCR recognition errors as mentioned in Section IV-A. In order to fairly compare with other state-of-the-art models, we disable the image extractor modules of VisualAES. The results are shown in Table III. We can see that VisualAES achieves the best performance with 0.855 Pearson Correlation and 0.786 Kappa Value. Note that the results of both Kappa and Pearson on ChnStd are considerably lower than ASAP, which could be attributed to smaller number of samples and dataset complexity. The improvements in both native and non-native datasets verify the progress of VisualAES and the result are from the designed ensemble module .

TABLE III
THE SINGLE-MODAL RESULTS ON CHNSTD DATASET. THE RESULTS OF VISUALAES IN THIS TABLE IS OBTAINED WITHOUT IMAGES IN ORDER TO COMPARING WITH OTHER MODELS, ONLY THE ENSEMBLE FUNCTION WORKS.

Model	Kappa	Pearson
Bert-base	0.768	0.838
Bert-large	0.621	0.730
Roberta-base	0.777	0.848
Roberta-large	0.501	0.540
Albert-base	0.739	0.803
Albert-large	0.772	0.809
VisualAES	**0.786**	**0.855**

E. Text-Image bi-modal Scoring

In the previous Section IV-C and IV-D, we evaluate VisualAES on both datasets with pure text inputs. In this section, we evaluate bi-modal VisualAES on the proposed ChnStd dataset by using both text and image as inputs. Moreover, in order to analyze the effect of incorporating bi-modalities, we also modify the previous pre-trained models (Bert, Roberta, Albert) to accept bi-modal inputs by adding a fully connected layer that transforms the combination of text and image into a vector. The results are shown in Table IV, we can see that with the help of images, VisualAES achieves a further improvement on ChnStd, with the highest 0.865 Pearson Correlation and 0.796 Kappa Value.

TABLE IV
THE BI-MODAL RESULTS ON CHNSTD DATASET WITH HANDWRITTEN IMAGES. THE SUBSCRIPT img REFERS TO MODELS PERFORM ON BOTH TEXT SEQUENCE AND HANDWRITTEN IMAGES.

Model	Kappa	Pearson
Bert-base_img	0.715	0.729
Bert-large_img	0.689	0.713
Roberta-base_img	0.625	0.659
Roberta-large_img	0.668	0.712
Albert-base_img	0.651	0.678
Albert-large_img	0.722	0.706
VisualAES	**0.797**	**0.865**

F. Effect of OCR Errors

As an indispensable pre-processing step for handwritten AES, OCR may introduce extra loss caused by automatic recognition errors. To verify if the proposed VisualAES could reduce the effect of OCR errors, we perform experiments using both human corrected text and original OCR results as input. The text correction is conducted by professional school teachers, and all students' spell mistakes are kept as well as unclear writings. The unchecked OCR outputs keep the recognition errors without any correction. The Accurate Rate (AR) [44] of OCR in ChnStd is 83.87%.

The results are shown in Table V. We can see that the OCR errors do decrease the performance of all models compared with Table III in terms of both Pearson and Kappa value. However, with the help of calligraphy images, the proposed VisualAES could gain more information from this extra modality to offset the negative effect of OCR errors in a certain content, and perform better than other models.

TABLE V
THE RESULTS ON UNCHECKED OCR DATA.

Model	Pearson	Kappa
bert_base_unchecked	0.759	0.754
roberta_base_unchecked	0.738	0.737
albert_base_unchecked	0.716	0.706
VisualAES_uncheckd	**0.762**	**0.756**

G. Generalization Capability

Generalization ability is vital for AES systems, since essays are usually from different students with various prompts. To evaluate the generalization capability of our VisualAES, we perform a training process on Grade 7 and test on Grade 9 essays. This is challenging since training and testing sets are collected from totally different students with various English abilities and prompts, and scored by different raters. That is, the model has to score new essays without any prior knowledge.

The results are shown in Table VI. It can be shown that all models suffer from considerable performance degradation. This could be attributed to sharp reduction of training samples and different scoring standards between Grade 7 and 9 teachers. However, the proposed VisualAES still yields the best results, especially the 0.58 Pearson correlation coefficient. This demonstrates great generalization capability of the VisualAES, with superior bi-modal fusion mechanism and stacking ensemble architecture.

TABLE VI
THE RESULTS OF GENERALIZATION TEST. ALL MODELS ARE TRAINED ON GRADE 7 AND TESTED WITH GRADE 9 DATASET.

Model	Pearson	Kappa
bert-base	0.476	0.179
roberta-base	0.517	0.044
albert-base	0.478	0.208
VisualAES	**0.58**	**0.241**

V. CONCLUSION

This paper proposes a novel bi-modal VisualAES system, which achieves outstanding performance for both typical and handwritten essay scoring by merging the image with text sequence. We also incorporate a stacking ensemble to further enhance its performance. The whole system is end-to-end, simple and effective. Experimental results prove that the VisualAES outperforms all existing methods on both the public ASAP and the proposed ChnStd datasets, which are designed for native and non-native speakers' assessment respectively. More importantly, by incorporating handwritten features, the VisualAES achieves a further performance improvement on ChnStd with the highest 0.865 Pearson Correlation and 0.796 Kappa Value. In addition, we also demonstrate that the bi-modal VisualAES could reduce the negative effect of OCR errors and benefit the generalization ability. One interesting direction for further research is to explore cross-domain or cross-language AES tasks using the latest pre-trained models and multi-modal information.

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